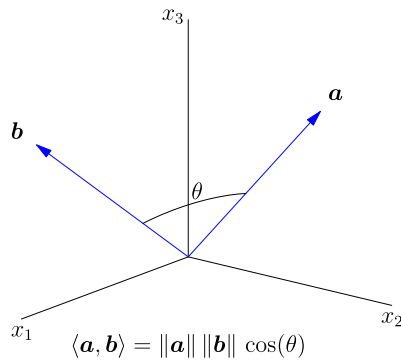




Inner products, operators

Recap

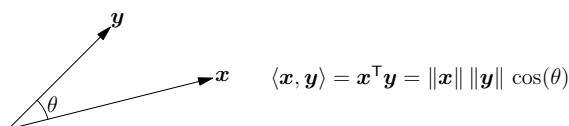
- We have looked at vector space (closed sets where we can add elements and multiply them by a scalar)
- Recall that vector spaces don't just apply to normal vectors ($\mathbb{R}^n$), but to matrices, functions, sequences, random variables, . . .
- Proper distances or metrics, $d(\mathbf{x}, \mathbf{y})$, allow us to construct ideas about geometry of the vector space
- Norms, $\|\mathbf{x}\|$, that allow us to reason about the size of vector
- Norm induce a distance, $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$

Axioms of Inner Products

- An inner product satisfies
 1. $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$ for all $\mathbf{x} \in \mathcal{V}$
 2. $\langle \mathbf{x}, \mathbf{x} \rangle = 0$ if and only if $\mathbf{x} = \mathbf{0}$
 3. $\langle \alpha \mathbf{x}, \mathbf{y} \rangle = \alpha \langle \mathbf{x}, \mathbf{y} \rangle$
 4. $\langle \mathbf{x}, \mathbf{y} + \mathbf{z} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{x}, \mathbf{z} \rangle$
 5. $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$
- We can show that $\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}$ satisfies the axioms of a norm, so that an inner-product space is a normed space
- The norm associated with the inner-product for vectors in $\mathbb{R}^n$ (i.e. $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y}$) is the Euclidean norm $\|\mathbf{x}\| = \sqrt{\mathbf{x}^\top \mathbf{x}}$

Angles Between Vectors

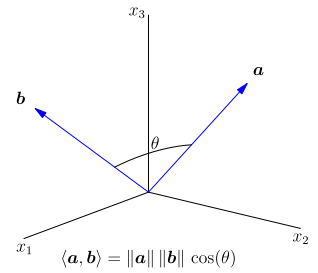
- A natural interpretation of the inner product is in providing a measure of the angle between vectors



- Vectors are orthogonal if $\langle \mathbf{x}, \mathbf{y} \rangle = 0$
- We can extend this idea to functions

$$\langle f(x), g(x) \rangle = \int_{x \in \mathcal{I}} f(x)g(x)dx = \|f(x)\| \|g(x)\| \cos(\theta)$$
- Note that $\sin(x)$ and $\cos(x)$ are orthogonal in the interval $[0, 2\pi]$

1. Inner Products
2. Operators



Inner Products

- We will often consider objects with an *inner product*
- For vectors in $\mathbb{R}^n$

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y} = \sum_{i=1}^n x_i y_i$$

- For functions

$$\langle f, g \rangle = \int_{x \in \mathcal{I}} f(x)g(x)dx$$

- For $m \times n$ matrices

$$\langle \mathbf{A}, \mathbf{B} \rangle = \text{Tr} \mathbf{A}^\top \mathbf{B} = \sum_{i=1}^m \sum_{j=1}^n A_{ij} B_{ij}$$

Cauchy-Schwarz Inequality

- One of the most important results of inner-product spaces, known as the **Cauchy-Schwarz inequality** is that

$$\langle \mathbf{x}, \mathbf{y} \rangle^2 \leq \langle \mathbf{x}, \mathbf{x} \rangle \langle \mathbf{y}, \mathbf{y} \rangle = \|\mathbf{x}\|^2 \|\mathbf{y}\|^2$$

- Or

$$|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \|\mathbf{y}\|$$

- This is a very general result so for example

$$\left| \int f(x)g(x)dx \right| \leq \sqrt{\left(\int f^2(x)dx \right) \left(\int g^2(x)dx \right)}$$

Basis Functions

- Any set of vectors $\{\mathbf{b}_i | i = 1, \dots\}$ that span the space can be used as a basis or coordinate system

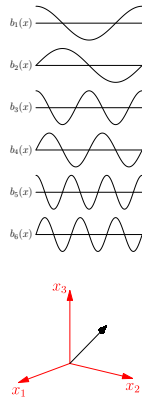
- The simplest and most useful case is when the vectors are orthogonal and normalised (i.e. $\|\mathbf{b}_i\| = 1$)

- In $\mathbb{R}^3$ we could use $\mathbf{b}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\mathbf{b}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, $\mathbf{b}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

- This is not unique as we can rotate our basis vectors

- For an orthogonal basis we can write any vector as $\hat{\mathbf{x}} = \begin{pmatrix} \mathbf{x}^\top \mathbf{b}_1 \\ \mathbf{x}^\top \mathbf{b}_2 \\ \mathbf{x}^\top \mathbf{b}_3 \end{pmatrix}$

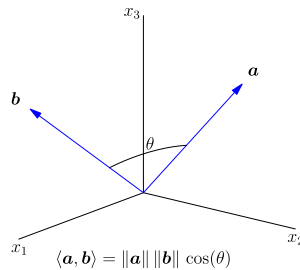
- For functions we can use any ortho-normal set of functions as a basis
- The most familiar are the Fourier functions $\sin(n\theta)$ and $\cos(n\theta)$
- Any function in $C(0, 2\pi)$ can be represented by a point $f = \begin{pmatrix} \langle f(x), b_0(x) \rangle \\ \langle f(x), b_1(x) \rangle \\ \vdots \end{pmatrix}$
- There might be an infinite number of components
- This is analogous to points in $\mathbb{R}^n$ (for large n)



- We have gone to these lengths as we want to show that many properties of vectors are shared by other objects (matrices, functions, etc.)
- The notions of distance (geometry), norms (size of vectors) and inner products (angles between vectors) provides a very rich set of concepts
- Vectors form the backbone of objects we will use repeated in machine learning
- The next piece of the jigsaw is to understand how we can transform these objects

Outline

- Inner Products
- Operators



Operators

- In machine learning we are interested in transforming our input vectors into some output predictions
- To accomplish this we will apply some mapping or operators on the vector $\mathcal{T} : \mathcal{V} \rightarrow \mathcal{V}'$
- This says that $\mathcal{T}$ maps some object $x \in \mathcal{V}$ to an object $y = \mathcal{T}[x]$ in a new vector space $\mathcal{V}'$
- This new vector space may or may not be the same as the original vector space
- Our objects may be any object in a vector space such as a function

Linear Operators

- Operators are in general very complicated, but a particular nice set of operators are linear operators
- $\mathcal{T}$ is a linear operator if
 - $\mathcal{T}[ax] = a\mathcal{T}[x]$
 - $\mathcal{T}[x + y] = \mathcal{T}[x] + \mathcal{T}[y]$
- For normal vectors ($x \in \mathbb{R}^n$) the most general linear operation is

$$\mathcal{T}[x] = Mx$$

where M is a matrix

Matrix multiplication

- For an $\ell \times m$ matrix A and an $m \times n$ matrix B we can compute the $(\ell \times n)$ product, $C = AB$, such that

$$C_{ij} = \sum_{k=1}^m A_{ik} B_{kj} \quad \left(\begin{pmatrix} \equiv \end{pmatrix} \begin{pmatrix} || \end{pmatrix} = \begin{pmatrix} \equiv \end{pmatrix} \right)$$

- Treating the vector x as a $n \times 1$ matrix then

$$y = Ax \Rightarrow y_i = \sum_j M_{ij} x_j \quad \left(\begin{pmatrix} \equiv \end{pmatrix} \begin{pmatrix} || \end{pmatrix} = \begin{pmatrix} \equiv \end{pmatrix} \right)$$

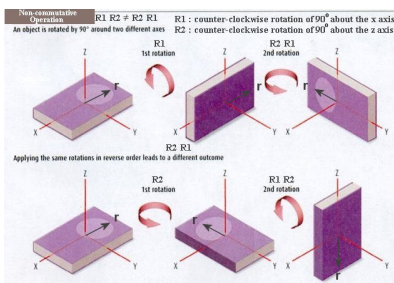
- Using the same matrix notation we can define the inner product as

$$\langle x, y \rangle = x^T y = \sum_{i=1}^n x_i y_i \quad \left(\begin{pmatrix} \longrightarrow \end{pmatrix} \begin{pmatrix} || \end{pmatrix} = \begin{pmatrix} \equiv \end{pmatrix} \right)$$

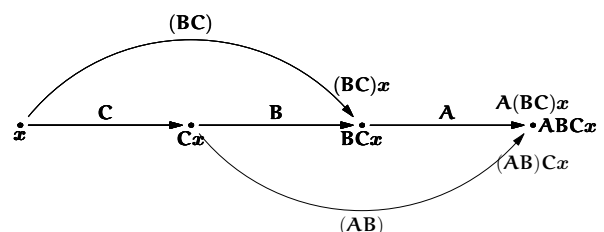
Non-commutativity

- In general $AB \neq BA$

$$\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$



Associativity of Mappings



- For all x we have $A(BC)x = (AB)Cx$
- This implies $A(BC) = (AB)C$

- The equivalent of a matrix for functions (i.e. a linear operator) is known as a kernel $K(x, y)$

$$g(x) = \mathcal{T}[f] = \int_{y \in \mathcal{I}} K(x, y) f(y) dy$$

- Our domain does not need to be one dimensional, e.g.

$$g(x) = \mathcal{T}[f] = \int_{y \in \mathcal{I}} K(x, y) f(y) dy$$

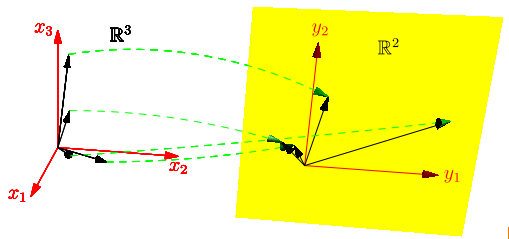
- We shall soon see examples of high-dimensional kernels

- Kernels are used heavily in machine learning
- In kernel methods such as SVM, SVR, Kernel-PCA
- They are also used in Gaussian Processes
- In all these cases we consider symmetric, positive semi-definite kernels
- Sometimes they can be interpreted as covariance between random functions

$$K(x, y) = \mathbb{E}_{f \sim \mathcal{P}}[(f(x) - \mu(x))(f(y) - \mu(y))]$$

General Linear Mappings

- In general a linear operator will map vectors between different vector spaces
- E.g. $\mathbb{R}^3 \rightarrow \mathbb{R}^2$



Summary

- We haven't covered much machine learning as such—sorry
- But mathematics is the language of machine learning and you have to get used to it
- Mathematics is like programming, if you don't understand the syntax and you can't write it down then it's meaningless
- We've taken a high level view of inner product spaces and operator, this will pay us back later as we look at kernel methods

Square Matrices

- We will spend a lot of time on operators that map from a vector space onto itself $\mathcal{T} : \mathcal{V} \rightarrow \mathcal{V}$
- For vectors in $\mathbb{R}^n$ such linear operators are represented by square matrices
- When there is a one-to-one mapping then we have a unique inverse
- We will study such mappings in detail in the next lecture